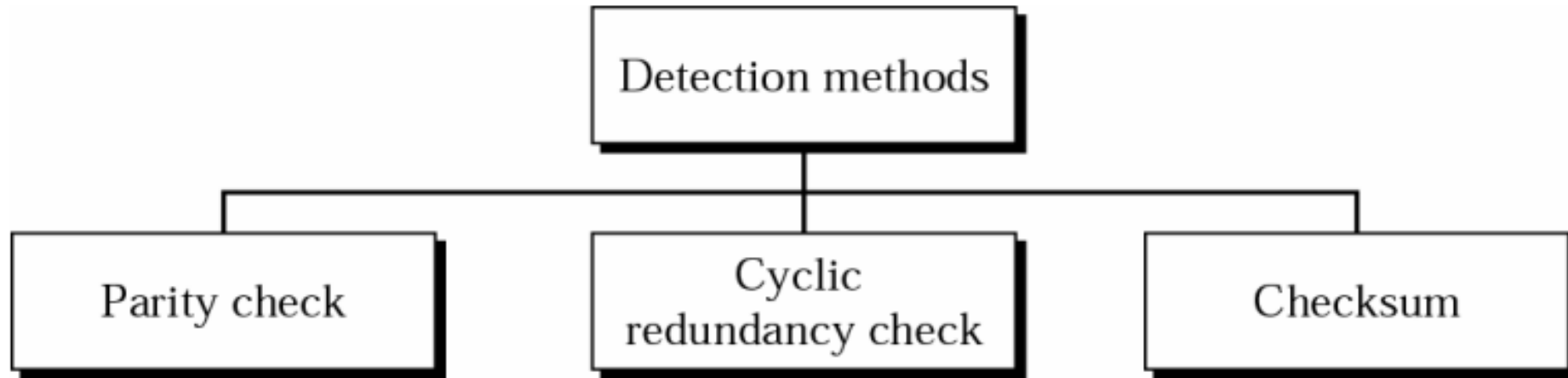
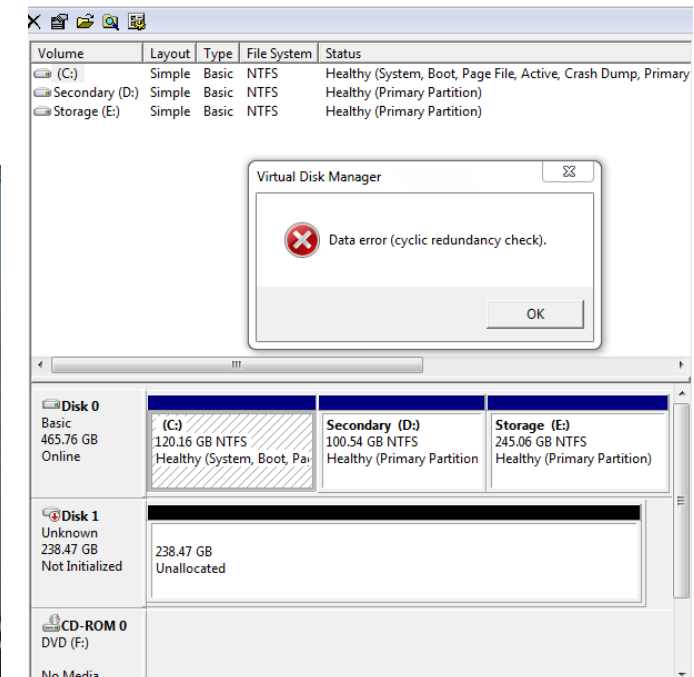
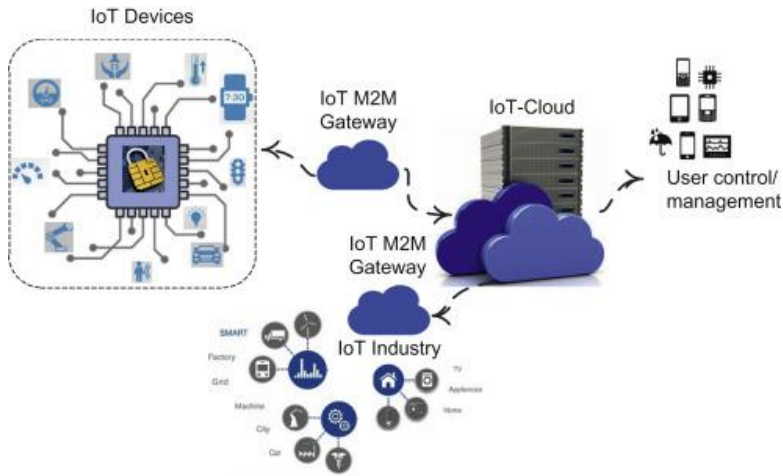


Detection Methods



Cyclic Redundancy Check (CRC) Methods

- More powerful error-detection coding.
 - One of the most commonly used method for error detection in digital networks and storage devices.
- 😊 Simple to implement and highly effective.



Key Elements of CRC

- 1. Data Block (Message):** The actual data in binary.
- 2. Predefined Divisor:** The protocols that the sender and the receiver mutually agree upon (=will be known to the sender and the receiver).
- 3. Remainder:** The result of the binary division appended to the message and transmitted.
- 4. CRC Code (Checksum):** The remainder appended to the original data to detect errors.

XORing of two single bits or two words

$$0 \oplus 0 = 0 \qquad 1 \oplus 1 = 0$$

a. Two bits are the same, the result is 0.

$$0 \oplus 1 = 1 \qquad 1 \oplus 0 = 1$$

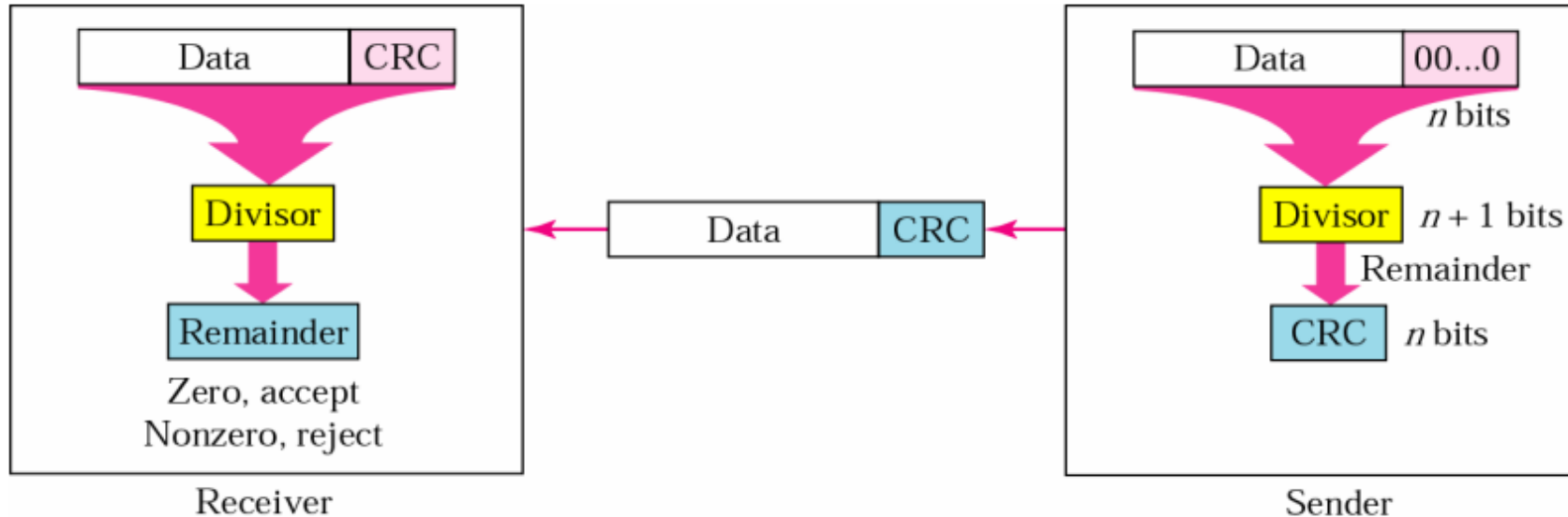
b. Two bits are different, the result is 1.

	1	0	1	1	0
$\oplus$	1	1	1	0	0
	0	1	0	1	0

c. Result of XORing two patterns

How CRC works

- **Sender:** CRC is generated (by calculation), resulting in redundant bits (CRC checksum) that are appended to the original data.
- **Receiver:** If the remainder is zero, accept the data; if the remainder is nonzero, reject the data.



CRC Calculation

1. Convert the Data in Binary (D).
2. Find the length (L) of the divisor (ตัวหาร) (G).
3. Append 0 equals “L-1” bits to D.
4. Perform binary division (D/G) operation using **XOR**
(Results in the quotient (ผลลัพธ์) and the remainder (เศษเหลือ))
5. **Remainder of the division = CRC** (ignore quotient).

Note The CRC must be of “L-1” bits.

Example 1: CRC Calculation (I)

1. Find the CRC for the data blocks 100100 with the divisor 1101?

Sender

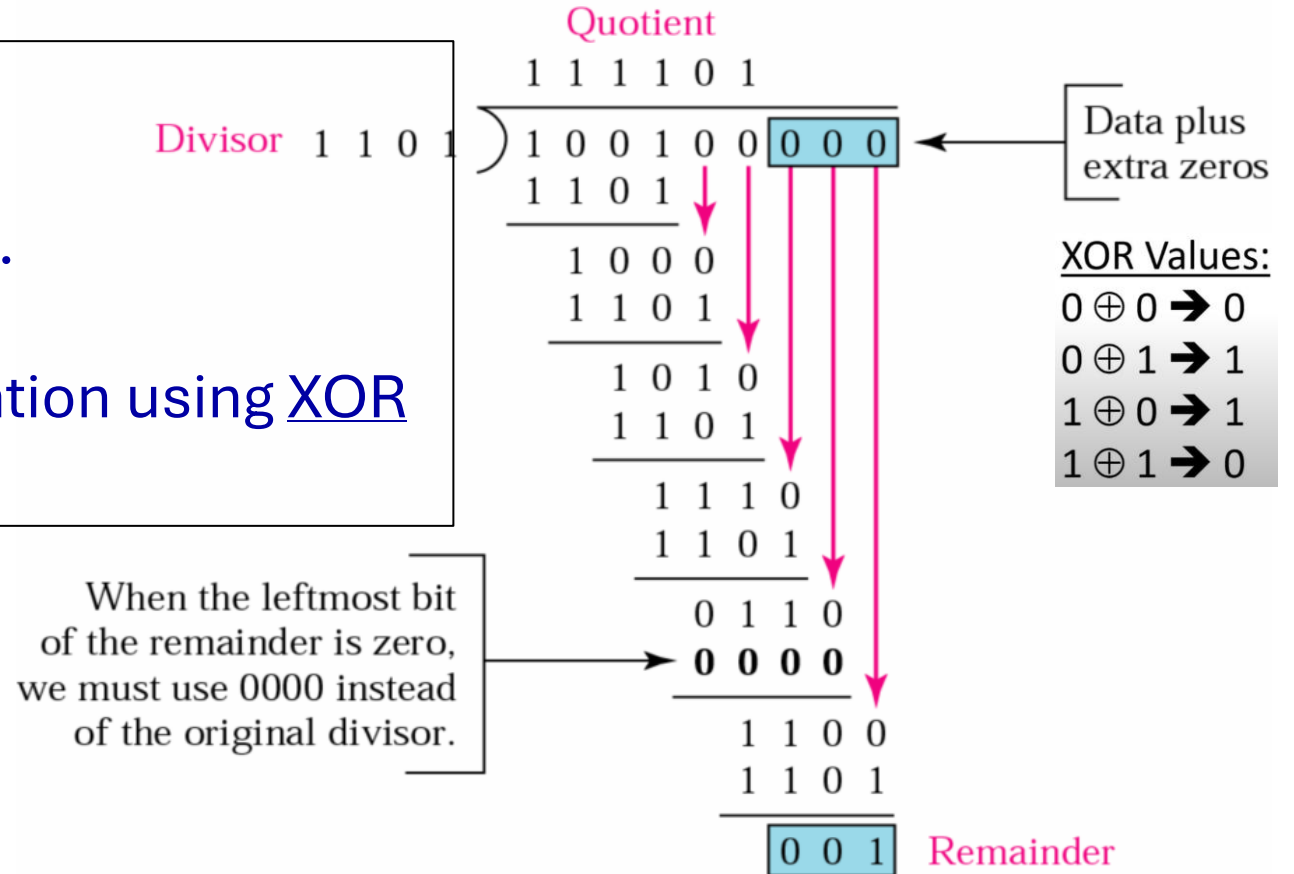
CRC calculation steps

1. Convert the Data in Binary (D).
2. Find the length (L) of the divisor (G).
3. Append 0 equals L-1 bits to D.
4. Perform binary division (D/G) operation using XOR
5. Remainder of the division = CRC.

Answer

CRC = 001

Data+CRC = 100100001



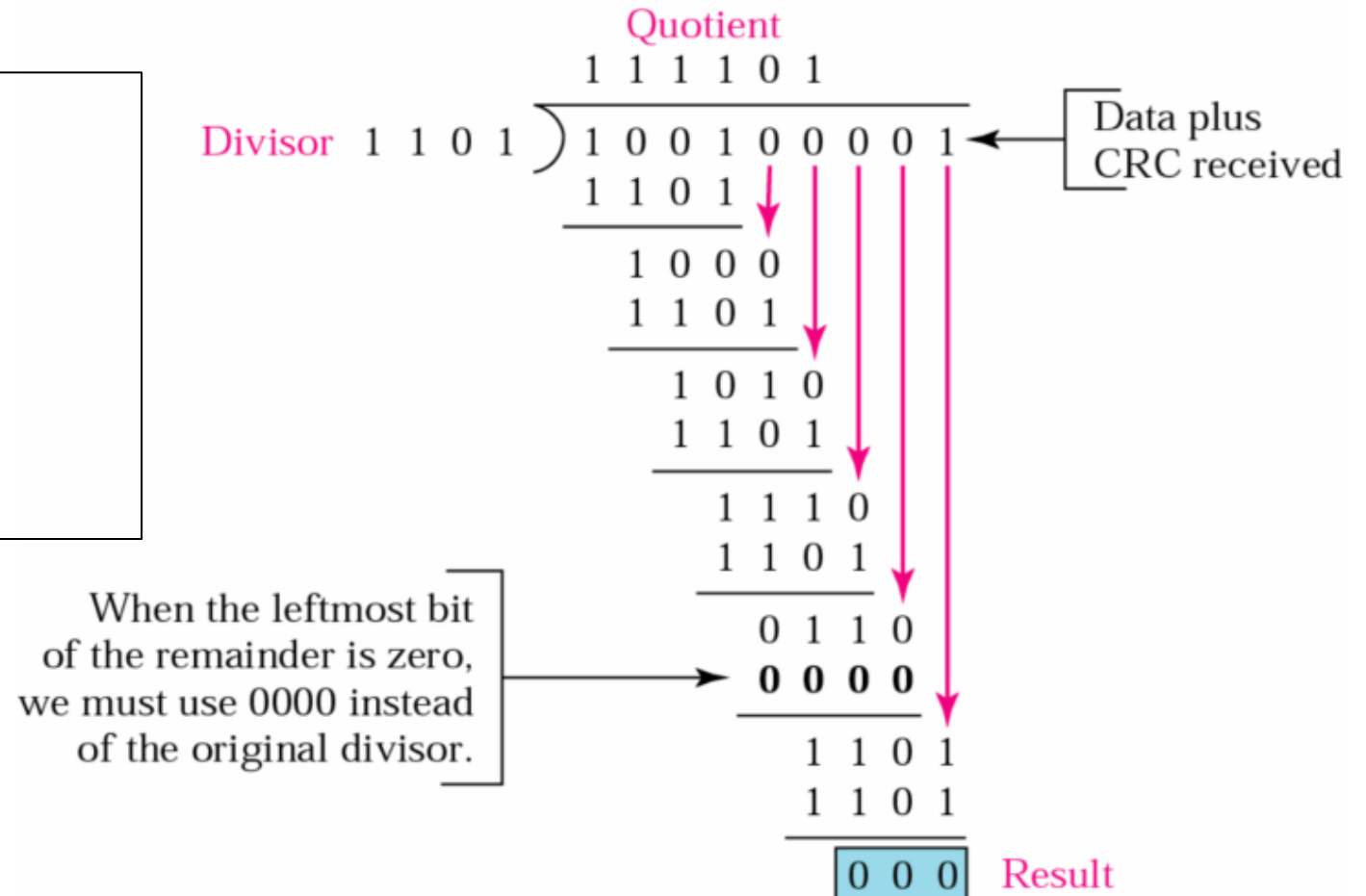
Example 1: CRC Calculation (II)

1. Find the CRC for the data blocks 100100 with the divisor 1101?

Receiver

CRC Verification

1. Binary division: Data = 100100001, Divisor = 1101
2. Remainder = **Zero** -> accept data, Remainder = **Nonzero** -> reject data



Divisor Polynomial

- The divisor in a CRC is a polynomial (พหุนาม)

- ☺ Convenient to represent binary data.

- ☺ Effective way to perform error-checking operations.

- ❖ E.g., $x^3 + x + 1$ can be represented as 1011.

$$x^7 + x^5 + x^2 + x + 1$$

Ideal properties of a CRC divisor

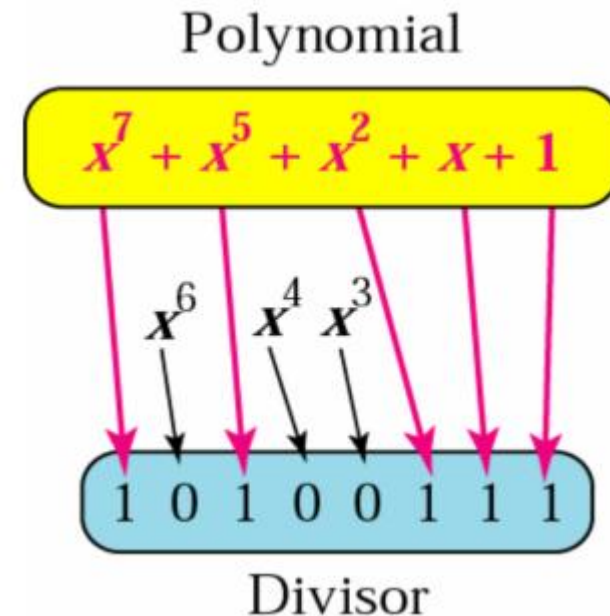
1. Must Not Be Divisible by (x)

-> Ensures detection of errors in the first bit.

2. Must Be Divisible by $(x + 1)$

-> Ensures detection of errors in the last bit.

3. Can Detect Burst Errors with a length $\leq$ degree of the polynomial.



Example 2: CRC Calculation

2. Find the CRC for the data blocks 11001001 with the divisor x^3+1

Sender

CRC Calculation Steps

1. Convert the Data in Binary (D).
2. Find the length (L) of the divisor (G).
3. Append 0 equals L-1 bits to D.
4. Perform binary division (D/G) operation using XOR
5. Remainder of the division = CRC.

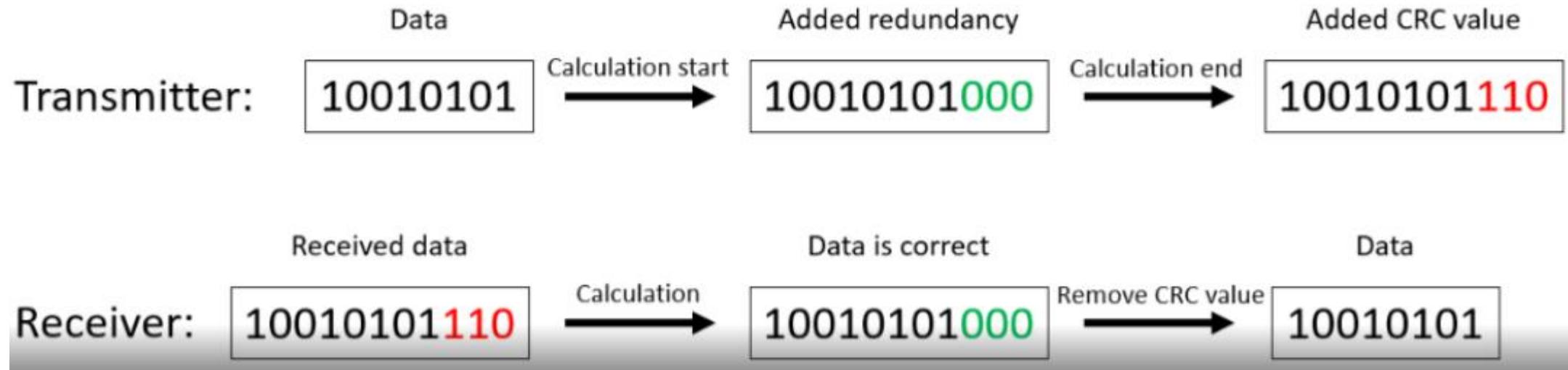
Answer

CRC = 011

Data+CRC = 11001001011

CRC Summary

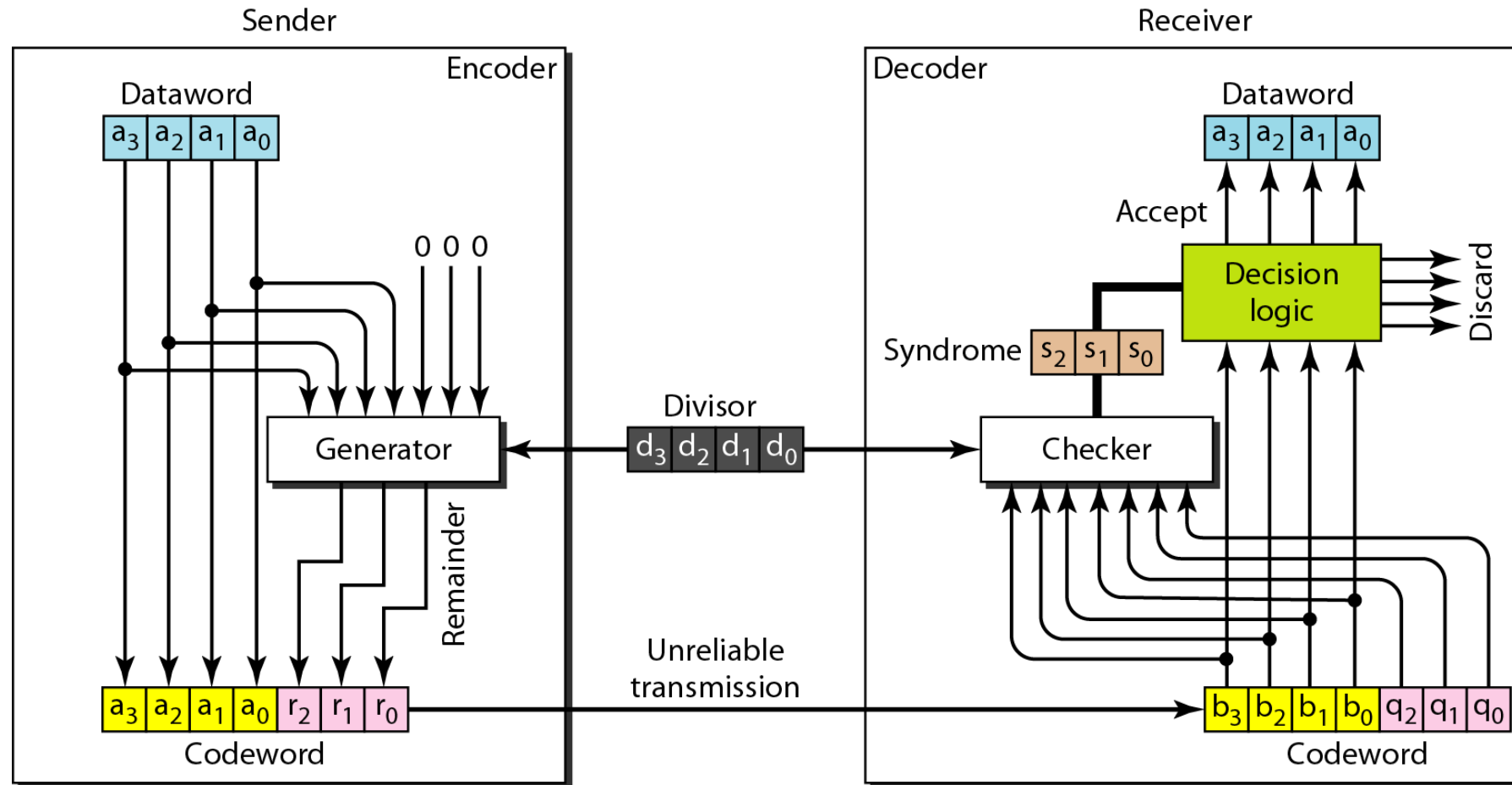
Applications of CRC include network communications e.g., Ethernet, IEEE 802.11 Wi-Fi), data storage devices (e.g., hard drives), and file formats (e.g., ZIP archives), USB (for token and data packets), HDLC (High-Level Data Link Control), PPP (Point-to-Point Protocol), and CAN bus (Controller Area Network used in automobiles).



Quiz: CRC Calculation

1. Find the CRC for the data blocks 11001001 with the divisor x^3+1
2. Find the CRC for the data blocks 1110010101 with the divisor x^3+x^2+1 . And perform division operations at the receiver.
3. Find the CRC for the data block 1010101010 using the divisor x^5+x+1 . And perform division operations at the receiver.
4. Find the CRC for the data block 1010001101 using the divisor $x^5+x^4+x^2+1$. And perform division operations at the receiver.

CRC encoder and decoder



Divisor (generator polynomial) ไม่ได้ถูกส่งไปพร้อมกับข้อมูล ถูกกำหนดไว้ในมาตรฐาน (**protocol specification**) เช่น Ethernet / IEEE 802.3 ใช้ CRC-32 polynomial, CAN bus (รถยนต์) ใช้ CRC-15 polynomial, USB ใช้ CRC5, CRC16

Division in the CRC decoder for two cases

